



Instructor: ..... Date: .....

Learner: ..... Regional Branch: .....

Program: ..... Center: .....

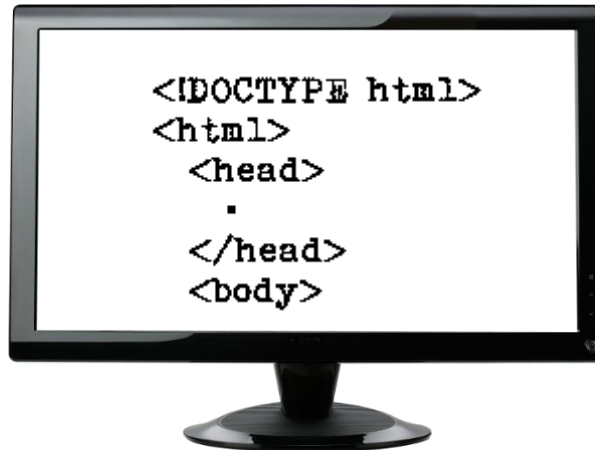
## CONTENT

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### Let's Start!

#### Exception Handling and Its Keywords in C#



#### Learning Outcome:

Talk about principles, concepts, characteristics of general-purpose computer programming language in spoken and written situations.

#### Ask your partner:

- What happens when an error occurs in the running of a code? How does the end user know what to do?
- What do you do when coding to foresee possible errors?

#### Can Do Statements:

- I can talk about the concept of exception handling in C#.
- I can discuss information about the keywords used for exception handling in C#.
- I can recall ideas about some exception classes in C#.



### Let's Explore!

Carl gives a lesson on exception handling in C#.



*Instructions: Read the text below.*

Hello, everyone. In today's session, we will talk about the concept of exception handling in C# and some related topics. At the end of this session, you will be able to manage a few keywords necessary to implement exception handling in your code. Exception handling is the method to face any error that can appear during the execution of a program or application. When you write code, you have to foresee any possible error during its execution process because when any occurs, the end user will need to know what is going wrong. To do this, the first keyword you have is "try." You have to enclose functions within this class having in mind that any error could occur with the operations inside it; someone could delete a file, change its name, etc.

Then, you have the keyword "catch." You can have one, a few, or many "catches" depending on the number of operations you have in each class. With this keyword, you will be able to set the reaction of the program or application each time an error appears; it could, for example, send a text to the end user with a few steps to check and recommend a possible solution. The third keyword we have is "finally." With this keyword, you define that whether an error appears or not, the operations run up to the result needed. In the end, you can use the keyword "throw." With this keyword, you will be able to send a new action through the code to verify no error from a chain of operations is left with no solution.

To finish our session, I'd like you to remember a few exception classes: `System.IO.IOException`, `System.DivideByZeroException`, and `System.OutOfMemoryException`. The first one handles errors connected to I/O properties; the second one, the errors coming from dividing with zero, and the third one, handles errors resulting from insufficient free memory.





### Let's Practice!

Steve will be able to understand this new concept.

*Instructions: Read the paragraph. Underline all the mistakes in the paragraph.*

An exception in C# is a problem in the execution of a program. From now on, Steve, a student who was in the C# lesson, will been able to implement exception handling when writing his codes. He was be able to work with keywords such as "try," "catch," and "finally." Steve will also be label to use the keyword "throw" to confirm that no mistake in the process chain is forgotten.



### Let's Practice!

Keywords in C#

*Instructions: Select the correct answer.*

1. This is the keyword you use for catching exceptions.
  - a. Try.
  - b. Catch.
  - c. Finally.
2. The keyword \_\_\_\_\_ allows the program run even whether the exception appears or not.
  - a. Try
  - b. Catch
  - c. Finally
3. The keyword \_\_\_\_\_ re sends the exception to confirm it is solved.
  - a. Finally
  - b. Throw
  - c. Try



## Extension Activity

Creating my own keywords in C#.

*Instructions: Complete the following steps. Fill in the Self-Assessment.*

- Think about the reason why those keywords were assigned to C# exceptions and write them down.
- Recall ideas about some exception classes in C#
- Create your own keywords for exception handling in C#.
- Explain what things will your keywords be able to do.
- Share and discuss with your partners and instructor.

### Self-Assessment

Yes

No

- I write the reason why those keywords were assigned to C# exceptions.
- I create my own keywords for exception handling in C#.
- I explain what the keywords will be able to do.
- I discuss information about keywords used for exception handling in C#

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## Target Words

- **face** (verb): to confront  
e.g. Exception handling is the method to face any error that can appear during the execution of a program or application.
- **foresee** (verb): to anticipate; to come up with a hypothetical future event  
e.g. When you write code, you have to foresee any possible error during its execution process because when any occurs, the end user will need to know what is going wrong.
- **I/O** (noun): input and output  
e.g. The first exception class handles errors connected to I/O properties.
- **enclose** (verb): to surround  
e.g. You have to enclose functions within this class having in mind that any error could occur with the operations inside it
- **within** (verb): in or into something else  
e.g. You have to enclose functions within this class having in mind that any error could occur with the operations inside it.



## Learning Points

### Will Be Able to Talk About Future Ability or Possibility

You use **be able to** for general ability and possibility.

You use **will be able to** to talk about possibility or ability in the future. **Will be able to** works as the future form of the verb **can** due to the impossibility of mixing the two verbs **will** and **can** in a single sentence.

Example:

- *You will be able to handle exceptions in C#.*

**Structure:**

| Subject | will | be able to | infinitive           |
|---------|------|------------|----------------------|
| Ruth    | will | be able to | explain the concept. |
| Baxter  | will | be able to | help.                |

#### Examples

- *At the end of this session, you will be able to manage a few keywords necessary to implement exception handling in your code.*
- *With this keyword, you will be able to set the reaction of the program or application each time an error appears.*
- *With this keyword, you will be able to send a new action through the code to verify no error from a chain of operations is left with no solution.*

### Quantifiers, a Few

You use quantifiers when you want to express the amount of something.

You use **few** or **a few** with plural countable nouns.

When you use **a few**, it means a small amount but enough. On the contrary, when you use **few**, it means not enough or too little.

Example:

- *A few employees have their presentations ready today.*

**Note:** **few** can have a positive meaning when it is followed by a noun that we don't want.

Example:

- *I have few problems understanding the concept of exception.*

#### Examples

- *At the end of this session, you will be able to manage a few keywords necessary to implement exception handling in your code.*
- *You can have one, a few, or many "catches" depending on the number of operations you have in each class.*
- *Send a text to the end user with a few steps to check and recommend a possible solution.*

